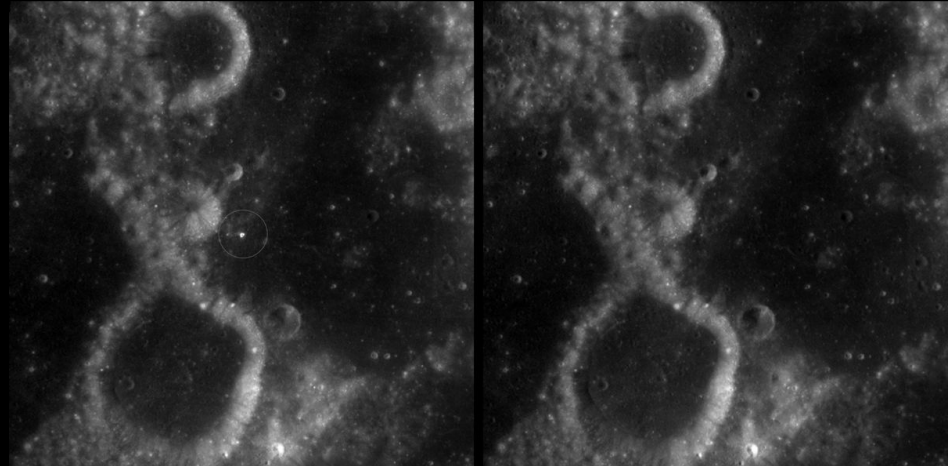


Announcements

- Weather permitting, we will go to the roof for observations after this introduction
- Next week is another night lab, here in PSF 647 - it will be the Intro to Telescopes Lab
 - You will need to print out the lab packet, as it is not a quiz in Canvas like the others we have done
- New crater on the Moon!
 - Formed between 4/11 and 5/22 2024
 - Named McGetchin
 - 728 feet wide and 141 feet deep

<https://science.nasa.gov/solar-system/moon/nasas-moon-orbiter-spots-new-once-in-century-moon-crater/>



Intro to the Night Sky



The Night Sky from a Cultural and Historical Perspective

- Observations of the night sky go to the earliest parts of human history
- The patterns we see have inspired and continue to inspire significant amounts of our culture and storytelling
 - Sometimes, the same sky-patterns appear in vastly different cultures, sometimes with different stories and names, and sometimes shared

Ursa Major and *Ursa Minor*, the “Great Bear” and “Little Bear” are said to be the nymph Callisto and her son (by Zeus) Arcas



They were sent to the stars to prevent a hunting accident.

Hera punished them by never allowing them to rest in the sea - so they rotate around the North Star and never set (at high enough latitudes - Tempe is right on the edge)

The word “galaxy” comes from the Greek for the Milky Way - *Galactica*



Battlestar Galactica sounds a lot cooler than *Battlestar Milky Smudge*



Taygeta

Asterope

Caleano

Electra

Maia

Merope

Alcyone

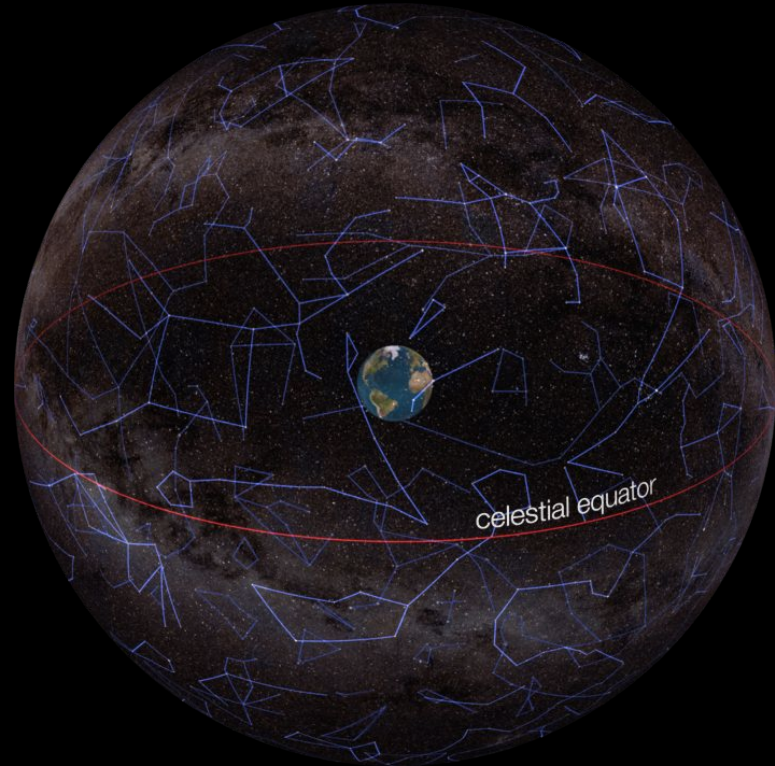
Pleione

Atlas

And now from a scientific perspective

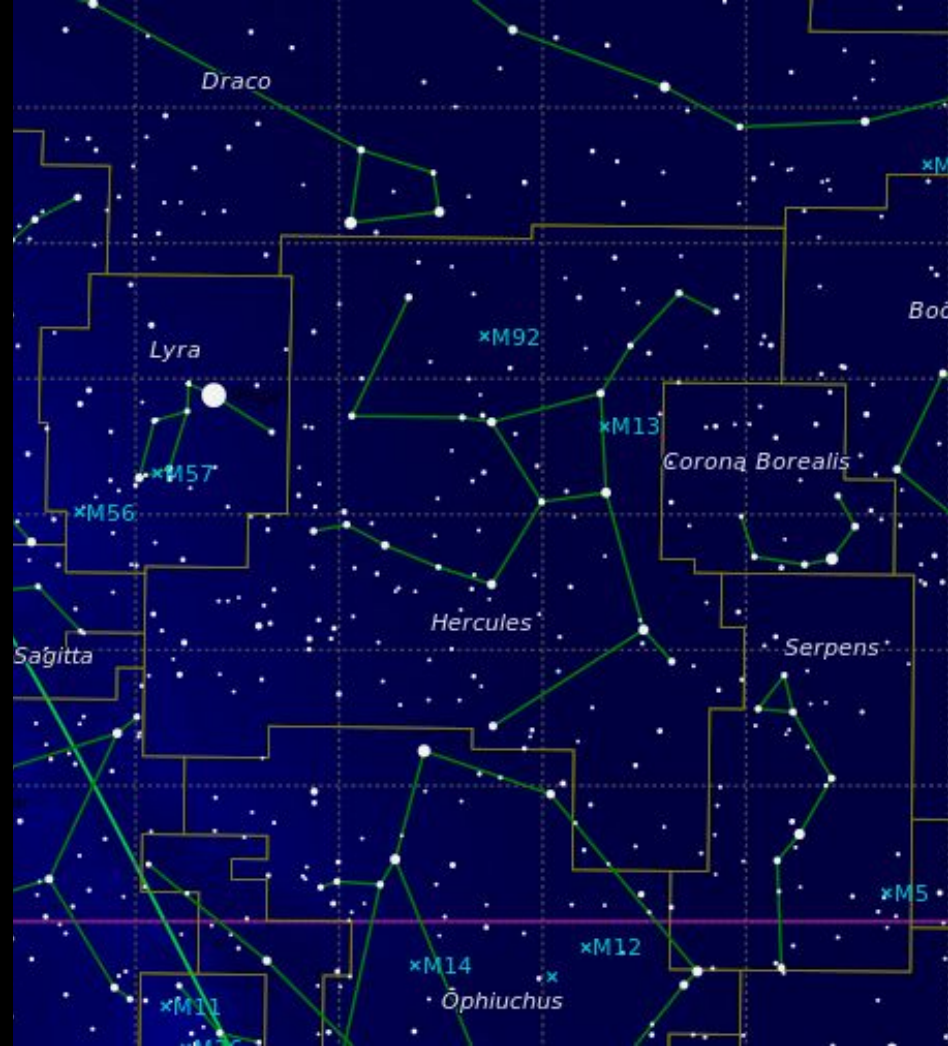
The Celestial Sphere and Celestial North Pole

- Think of the sky as a sphere surrounding the Earth - the celestial sphere
 - We can map things onto this sphere
 - Think about how a circle is 360 degrees around - and a sphere is a 3D circle
 - So angular size is how many of those 360 degrees does something take up
- The North Celestial Pole (NCP) is where the Earth's north pole, projected outward into space, intersects the celestial sphere
- Polaris (the North Star) is very close to the NCP
 - Because the NCP is based on the Earth's axis of rotation, this means it barely moves over the course of a day



Constellations

- Definition: **Area** of the sky where a group of stars form a recognizable pattern
- Officially, there are 88 of them
- Every point in the sky is in one of them
- If an object like the Moon, a planet, or an unknown object is found within a constellation's borders - it is said to be “within” that constellation



Special Types of Constellations

- **Zodiacal Constellations**: 13* constellations that the Sun appears to pass through over the course of a year, as the Earth orbits it
 - This path the Sun takes is called the ecliptic - it also represents Earth's orbit
- The 12 Zodiacal Constellations you may be familiar with are: Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpius, Sagittarius, Capricornus, Aquarius, and Pisces
- There is also Ophiuchus, but we can blame the Babylonians for that not being one of the traditional ones anymore

Special Types of Constellations (Cont.)

- **Circumpolar Constellations**: Those close enough to the NCP that they never set
- Which constellations are circumpolar depends on your latitude - a higher latitude means more circumpolar constellations
 - At the equator (latitude = 0°) there are no circumpolar constellations, at the north pole (latitude = 90°) all visible constellations are circumpolar
 - Here in Tempe (latitude = 33.43°), all constellations within 33.43° of the NCP are circumpolar (not accounting for buildings or mountains blocking our view) - *Ursa Minor*, *Draco*, *Cepheus*, *Cassiopeia*, and depending on how strict you are with definitions; also *Camelopardalis* and *Ursa Major*

Asterisms

- We don't typically talk about entire constellations, especially in casual use
 - Instead, we talk about patterns within (or across) constellations and prominent groups of stars
- These are *Asterisms*
- Examples:
 - The Big Dipper (part of *Ursa Major*)
 - The Great Square of Pegasus
- The constellations are named after asterisms, but not all asterisms are constellations



Stellar Magnitudes and Brightness

- Apparent magnitude is how we describe how bright a celestial object appears to be
- The scale is an **inverted, logarithmic scale**, meaning **smaller numbers are brighter objects**
 - Another way to think of it is a ranking system - magnitude 1 are the brightest, magnitude 2 the next, etc. Then we had to add negative numbers...
- On star charts, magnitude is denoted by the size of the dot - brighter objects get bigger dots
- It is apparent magnitude, because it is partially dependant on how far away an object is
 - Not just the objects own brightness
 - Closer objects appear brighter

Mag. 1



Mag. 2 x 2.5 dimmer



x 2.5

Mag. 3 x 2.5 dimmer



x 6.25

Mag. 4 x 2.5 dimmer



x 16

Mag. 5 x 2.5 dimmer

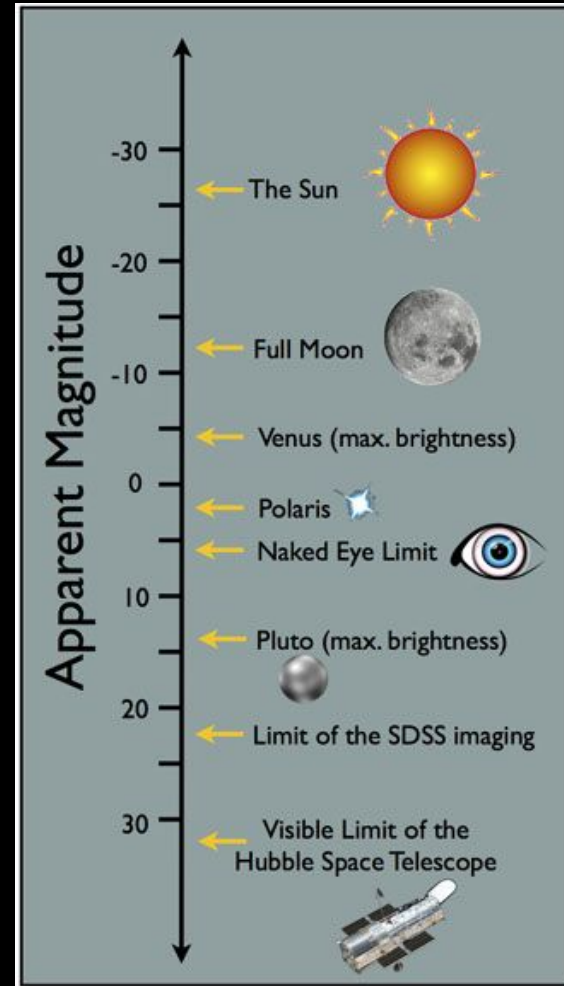


x 40

Mag. 6 x 2.5 dimmer



x 100



Today's Lab (Part 1)

- Do questions 1-3 now (basically a review of last week's lab) and let me know when you are done

Roof Rules (If Weather is good...)

- Bring your stuff with you
 - We will not be coming back to the classroom
- Do not enter the fenced in area(s)
- Red flashlights only
 - No white lights except in emergencies
- We will be up there with one or more other classes
 - You can ask me or the other TA(s) if you have any questions

Today's Lab (Part 2)

- If weather allows:
 - We will go to the roof
 - Make observations of the sky to complete your sketches
 - Then do the remaining questions (4-15)
- If weather is poor:
 - We will use Stellarium, you can open it on your own device, and I will put it on the board for you
 - Use this to complete your sketches, and then do the remaining questions (4-15)
- Questions 4-15:
 - You will first use the zodiacal constellations to find the ecliptic on a star chart and compare the predicted path of the unknown object
 - Then compare the unknown object's brightness over time to determine if it is coming towards or away from Earth

Sketching Tips

- Make an attempt at all sketches
 - I will not be grading you on artistic skill
 - As long as you make a genuine attempt and I can tell what you were trying to do, you will get full credit
- Draw what you can see
 - Not what you think we want you to see
 - Some of you may have better eyesight and can see more stars than others - draw them
 - If you can't, don't
 - If an object is blocked by a building or cloud - note that (or even draw said cloud/building)
 - Use the size of dots to represent brightness - the brighter an object, the bigger the dot
- The bottom of the page is the horizon, and the top is straight overhead

Image Credits (In order of appearance)

- NASA
 - <https://science.nasa.gov/solar-system/moon/nasas-moon-orbiter-spots-new-once-in-century-moon-crater/>
- Atlas Obscura
 - <https://www.atlasobscura.com/articles/what-color-is-night>
- Scouter Life
 - <https://www.scouterlife.com/blog/2023/1/31/is-the-big-dipper-a-constellation>
- Battlestar Galactica: Deadlock (Video Game)
 - <https://www.artstation.com/artwork/dyVImJ>
- Telescopic Watch
 - <https://telescopicwatch.com/how-to-find-pleiades-star-cluster/>
- Wikipedia - Celestial Sphere
 - https://en.wikipedia.org/wiki/Celestial_sphere
- Screenshot of <https://stellarium-web.org/>, taken by me
 - Location: Tempe, AZ and date/time: 2/10/26 at 21:16 local time
- EarthSky
 - <https://earthsky.org/astronomy-essentials/what-is-stellar-magnitude/>
- SDSS Voyages
 - <https://voyages.sdss.org/preflight/light/magnitudes/>